



JAIPUR NATIONAL UNIVERSITY

End Semester/Year Examination December -2025

Name of School:	School of Computer and Systems Sciences
Name of Programme / Course:	BCA / BCA (AIML) / BCA (Cyber Security) / PGDCA
Semester Year:	I / I
Paper Code:	BCAEMCO102T24
Title of Paper:	Elementary Mathematics
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section A and Section B. Section-A consists of seven questions and Section-B consists of six question. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	Section-A Attempt any five questions.	Marks Allotted	Bloom Level	Course Outcome
Q.1.	Analyzing If $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ verify that $A^2 - 4A - I = 0$.	4	L1	CO2
Q.2.	Solve the following. $\int (9x^2 + 6x + \frac{1}{x} + \sin 3x + \frac{1}{x^2}) dx$	4	L5	CO1
Q.3.	Find the Derivative of the functions w.r.t. x if $y = \frac{3\sin x}{x^4 + 1}$	4	L6	CO3
Q.4.	Compose the vertex, focus, directrix, axis, length of the latus rectum of parabola $y^2 = 4x$.	4	L2	CO4
Q.5.	if $A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 3 \\ 10 & 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 4 & 5 \\ 1 & 1 & 0 \end{bmatrix}$ find the value of AB.	4	L3	CO3

Q.6.	Explain the six different types of matrix.	4	L5	CO6
Q.7.	Summarize $\lim_{x \rightarrow 1} \frac{50x^2 + 60x + 60}{16x^2 + 10x - 1}$	4	L4	CO4
Section-B				
Attempt any four questions.				
Q.1.	Explain the inverse of matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$	10	L1	CO6
Q.2.	Solve $\lim_{x \rightarrow 4} \frac{x^2 - 10}{\sqrt{x^2 + 9} - 10}$	10	L2	CO3
Q.3.	Find the value of $\int \frac{4x^6 - 2x^2 + 5x + x^2 a^3}{x^2} dx$	10	L2	CO4
Q.4.	Describe the angle between the straight lines $x - \sqrt{3}y = 5$ and $x + \sqrt{3}y = 7$	10	L1	CO4
Q.5.	Using by matrix method find the value of x, y, z $\begin{aligned} x + 3y + 4z &= 8 \\ 2x + y + 2z &= 5 \\ 5x + y + z &= 7 \end{aligned}$	10	L5	CO2
Q.6.	Compose the second derivative or y'' if $y = xe^x + \cos 2x + \tan x + \sec x$.	10	L4	CO3